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Performance Evaluation and Mechanical Characteristics of Calcinated Red Mud-Based Cement for Eco-Friendly Surface Casing

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Abstract

The Bayer process for alumina production generates substantial quantities of red mud (RM), a highly alkaline industrial byproduct enriched with metal ions, presenting significant environmental management challenges. Approximately 1.5 tons of RM are produced per ton of alumina, contributing to global stockpiles estimated at nearly 4 billion tons. Conventional disposal methods, such as landfilling or aquatic dumping, are both economically burdensome and environmentally detrimental. This study explores the feasibility of incorporating calcinated red mud (CRM) as a partial cement replacement in lightweight cement slurries (LWS) for surface casing applications at depths of 3000 ft.

Two cement systems were formulated: a reference neat cement slurry (base slurry, BS) and a CRM-enhanced slurry. CRM was produced by calcining raw RM at 750 °C for 6 hours. Cement slurries were prepared in accordance with API standards and cured at 125 °F for 24 hours prior to testing. Mechanical and physical performance—including compressive strength, tensile strength, and porosity—was evaluated after 1 and 3 days of curing.

Initial results indicated a lower 1-day compressive strength for CRM (11.4 MPa) compared to BS (28.9 MPa). However, CRM exhibited a significant strength gain of 80% by day 3, reaching 20.5 MPa, whereas BS showed a decline to 24 MPa. Tensile strength results demonstrated consistent superiority of CRM over BS, with 20% and 3% higher values on days 1 and 3, respectively. Porosity measurements revealed slightly elevated porosity in CRM samples; however, a decrease from day 1 to day 3 suggests continued hydration and matrix densification.

These findings highlight the potential of CRM as a sustainable alternative material in LWS formulations for well cementing. To the best of our knowledge, this is the first study to systematically investigate the application of thermally treated RM in cementing operations, underscoring its promising role in advancing environmentally responsible cement technologies.

Keywords: Red mud, Cement formulation, Environment, Sustainability, Casing and cementing

Introduction

Cementing oil and gas wells involves placing cement slurry into the annular space behind the casing string to bond it with the borehole (Kiprotich Bett, 2010). This process is essential for maintaining well stability, ensuring productivity, achieving environmental compliance, and preserving subsurface integrity (Bois et al., 2012; Yousuf et al., 2021). The cement sheath performs multiple functions, including preventing fluid migration, securing the casing, resisting corrosive fluids and mechanical stress, and supporting wellbore walls (Rabia, 1985; Schlumberger, 2006; Plank et al., 2014).

Cementing operations are classified as primary (initial annular sealing) and secondary (remedial applications like plugging and squeezing) (Skadsem et al., 2024). The American Petroleum Institute (API) specifies oil well cement classes, with Class G and H being most common (Zhang et al., 2010). Additives are used to tailor slurry properties such as setting time, strength, density, and viscosity (Schlumberger, 2006; Broni-Bediako et al., 2016).

Cement slurry design is dictated by downhole conditions—pressure, temperature, and formation fracture gradient (Adjei and Elkatatny, 2021). In shallow formations, low fracture pressures necessitate the use of lightweight cement slurries (LWS) to avoid formation damage (Anyia, 2018). LWS offers a cost-effective alternative to multi-stage cementing but often requires performance optimization due to lower strength. Common LWS types include water-extended, foamed, and low-specific gravity systems, enhanced with additives like epoxy resins, microspheres, and silica fume (Schlumberger, 2006; Deshpande et al., 2019).

Recent studies have demonstrated effective LWS formulations with improved mechanical properties and reduced densities. Adjei et al. (Adjei et al., 2020) developed a slurry using 45% fly ash, 25% ground granulated blast furnace slag (GGBFS), and 5% silica fume by weight of blend (BWOB), achieving a density of 13.5 ppg. In a separate study, Adjei et al. (Adjei, Elkatatny, Sarmah, and Abdelfattah, 2021) replaced 10–30% of cement with calcined calcium bentonite (CCB) and incorporated 5–10% silica fume (by weight of cement, BWOC), producing a 12.5 ppg slurry. Adjei et al. (Adjei, Elkatatny, Sarmah, and Chineia, 2021) also evaluated dehydroxylated sodium bentonite at 10–50% BWOC, resulting in a stable slurry with a density of 13 ppg. Larki et al. (Larki et al., 2019) formulated an LWS with 30% natural pozzolan (BWOC), yielding a 14 ppg density. Kremieniewski (Kremieniewski, 2020b) utilized 15–60% cenospheres (BWOC), achieving densities ranging from 9.5 to 12 ppg, while a subsequent study by the same author (Kremieniewski, 2020a) demonstrated a 15% cenosphere-based system with a density of 12.9 ppg. These formulations highlight the potential of diverse additives—including pozzolanic materials and microspheres—for tailoring LWS to meet performance requirements under varying wellbore conditions.

Red mud (RM), a byproduct of the Bayer process for alumina production, poses significant environmental challenges due to its high alkalinity and metal content (Paramguru et al., 2004; Power et al., 2011; Wang et al., 2013, 2019; Archambo and Kawatra, 2021). Its pozzolanic reactivity and dicalcium silicate content make it a promising supplementary material for cement production, offering improvements in early strength and sulfate resistance without compromising mechanical integrity (Liang et al., 1996; Pan, 1999; Liu and Zhang, 2011; Qiu and Qi, 2011; Liu and Wu, 2012; Venkatesh et al., 2020). Prior studies have shown that RM, particularly when calcinated, can enhance hydration and strength in oil well cement systems, supporting its potential integration into LWS formulations for sustainable and high-performance applications (Liu and Poon, 2016; Cheng et al., 2021; Venkatesh et al., 2021; Mahmoud and Elkatatny, 2024).

This study evaluates the performance of calcinated red mud (CRM) as an additive in lightweight cement slurry (LWS), with a focus on the evolution of mechanical properties over time. LWS samples containing 15% CRM BWOC were prepared, along with a control sample without CRM for comparison. Mechanical testing was carried out at one and three days after preparation. The findings offer valuable insights into the potential of CRM as a sustainable and cost-effective additive for LWS in oil and gas well cementing applications.

Materials and Methods

Materials and characterization

Saudi Class G cement and the viscosifier were provided by TAQA Company, Saudi Arabia, while RM was sourced from MA'ADEN Aluminum Company, Saudi Arabia. RM underwent a calcination process involving heating at 750 °C for 6 hours to enhance its pozzolanic activity.

Both the cement and CRM were characterized using X-ray diffraction (XRD), X-ray fluorescence (XRF), and scanning electron microscopy (SEM) to evaluate their mineralogical and morphological properties. Additionally, particle size distribution (PSD) analysis was conducted for both materials to assess their granulometric characteristics.

Preparation

The two cement slurries were prepared in accordance with API procedures (API, 1991, API, 2013), with compositions detailed in Table 1. The first formulation, referred to as the base slurry (BS), consisted of Class G cement, viscosifier, and water. The second formulation incorporated 15% CRM by weight of cement (BWOC) as a partial replacement for cement. To maintain a consistent slurry density of 101 pcf, the water content was adjusted accordingly. The viscosifier concentration was fixed at 0.16% BWOC for both slurries. For clarity in result discussions, the formulations are abbreviated as BS for the base slurry and CRM for the calcinated red mud slurry.

Table 1—Details of the cement formulations prepared for this study.

Component	Wt. %	
	BS	CRM-based
Cement	500	425
CRM (15%)	-	75
Viscosifier (0.12%)	0.6	0.6
Water	391	381

Testing

Following sample preparation, all specimens were cured at 125 °F for 24 hours prior to testing. Mechanical and physical properties were evaluated at 1 and 3 days, including compressive strength using a scratch testing machine, tensile strength using a crushing machine, and porosity measurements via Nuclear Magnetic Resonance (NMR). All testing procedures adhered to API standards and recommended practices (API, 2013).

Results and Discussion

Characterization

The XRD analysis, summarized in Table 2, revealed distinct mineralogical compositions for cement and CRM. The cement primarily consisted of Tricalcium Silicate (C_3S , 65%), Dicalcium Silicate (C_2S , 15%), and a combined phase of Tetracalcium Aluminoferrite and Tricalcium Aluminate ($C_4AF + 2C_3A$, 18%), with Tricalcium Aluminate (C_3A) itself present at less than 1%. In contrast, the CRM composition was dominated by Cancrinite (71.7%) and Hematite (28.3%), indicating a significant transformation of the original RM phases during calcination. These results suggest that while cement is rich in silicate phases essential for strength development, CRM contains stable aluminosilicate and iron oxide phases that may influence hydration kinetics.

Table 2—XRD analysis of cement and CRM.

Cement	Wt. %
C ₃ S	65
C ₄ AF + 2C ₃ A	18
C ₂ S	15
C ₃ A	< 1
CRM	Wt. %
Cancrinite	71.7
Hematite	28.3

Elemental composition determined by XRF (Fig. 1) shows that CRM contains lower levels of Fe and Al than cement, whereas its Cl and Ca contents are significantly higher.

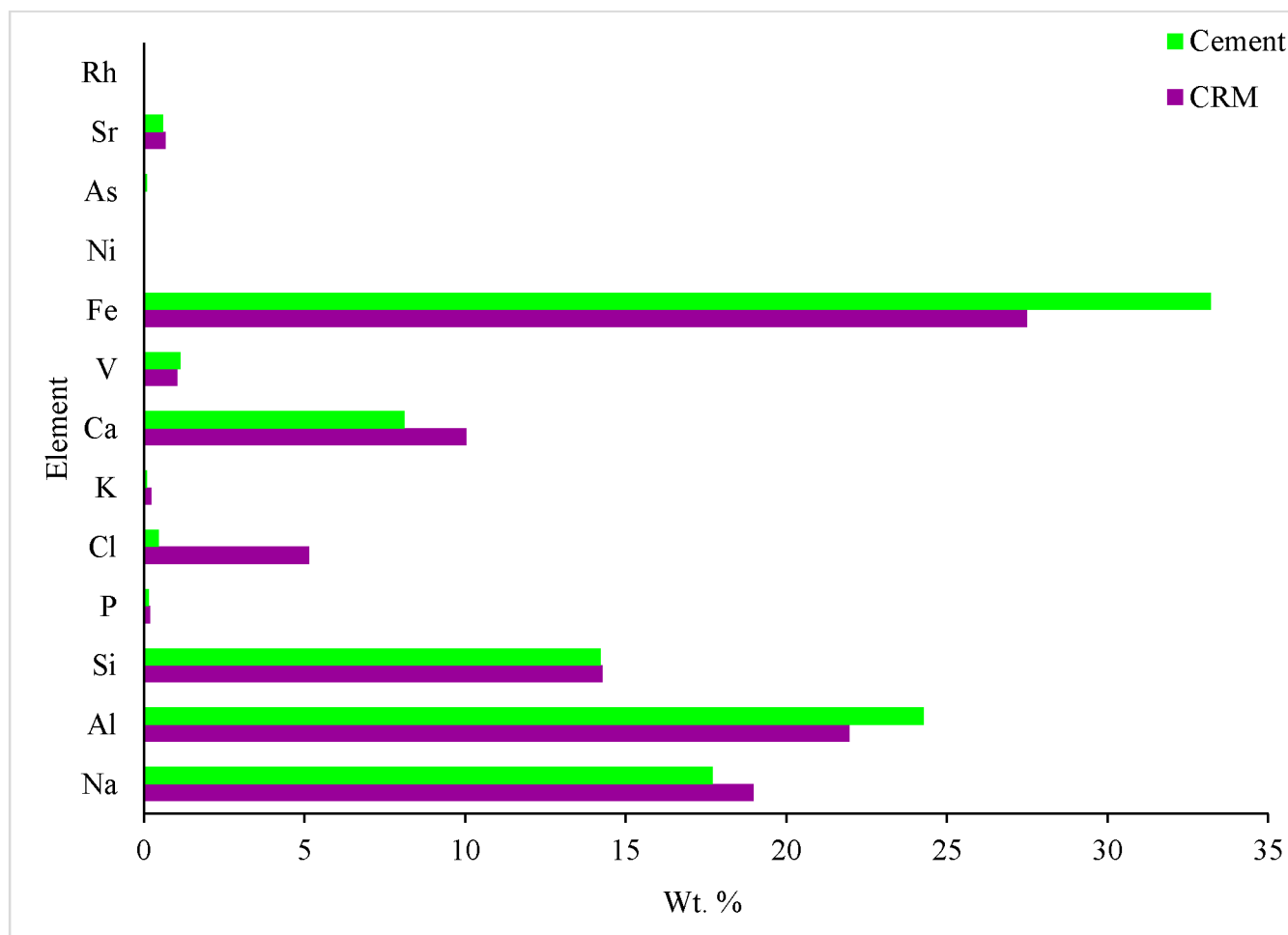


Figure 1—XRF analysis of cement and CRM.

SEM images (Fig. 2) reveal that both cement and CRM particles possess irregular shapes. Cement particles appear predominantly round with smooth surfaces, while CRM particles exhibit rougher textures, likely resulting from the high-temperature calcination process.

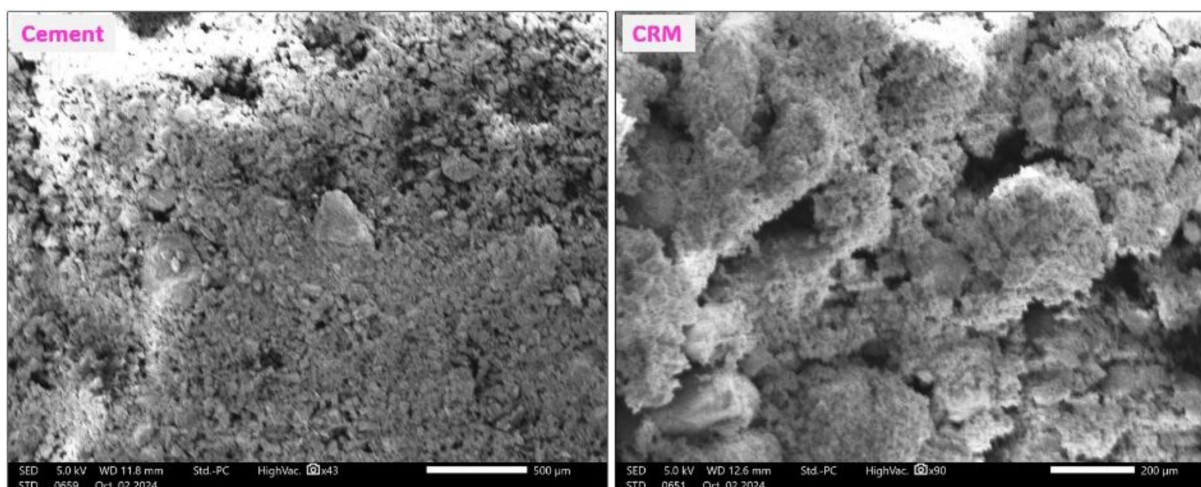


Figure 2—SEM imaging of cement and CRM.

PSD results for both samples are summarized in Fig. 3. The cumulative distribution function was utilized to determine the intermediate particle size (D_{50}). It was found that the corresponding D_{50} particle sizes for cement and CRM were 12 μm and 441 μm , respectively. This indicates that the CRM particles are significantly larger than the cement particles.

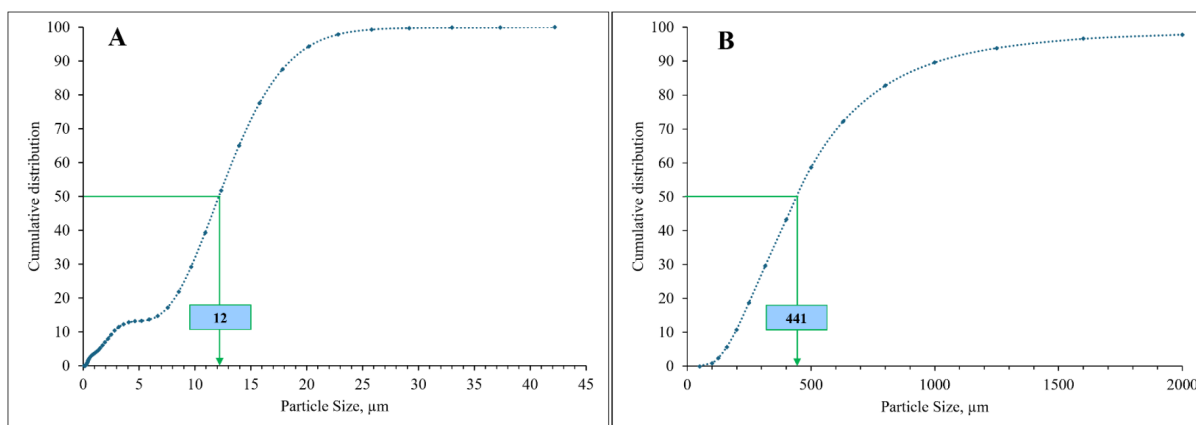


Figure 3—Particle size distribution of cement (A) and CRM (B).

Compressive strength

The compressive strength results obtained via scratch testing for all cement formulations are presented in Fig. 4. On the first day, BS exhibited a significantly higher compressive strength of approximately 29 MPa, compared to 11.4 MPa for the CRM-containing samples. However, after 3 days, the CRM sample demonstrated an 80% increase in strength, reaching 20.5 MPa, while the BS sample experienced a 17% reduction, dropping to 24 MPa. This progressive increase in compressive strength for the CRM formulation suggests a slower reaction rate among its constituents. Nevertheless, the strength gain over time indicates enhanced long-term stability, implying that CRM-based slurries may develop more durable strength characteristics with extended curing.

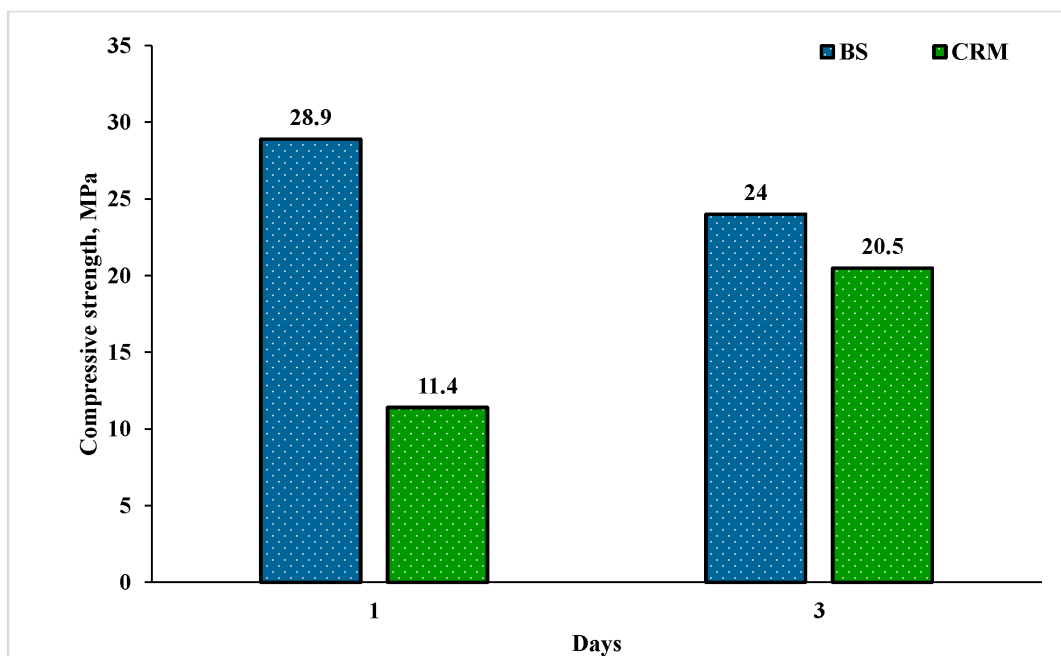


Figure 4—Compressive strength results from scratch tests for BS and CRM-based slurry.

Tensile strength

The tensile strength results for both cement formulations are shown in Fig. 5. Across all testing intervals, the CRM-based slurry demonstrated superior tensile strength compared to the BS. On day 1, the CRM formulation achieved a tensile strength of 1.09 MPa, approximately 20% higher than the BS sample (0.91 MPa). By day 3, both formulations exhibited considerable strength development, with CRM reaching 1.99 MPa and BS slightly lower at 1.93 MPa—a 3% difference in favor of CRM.

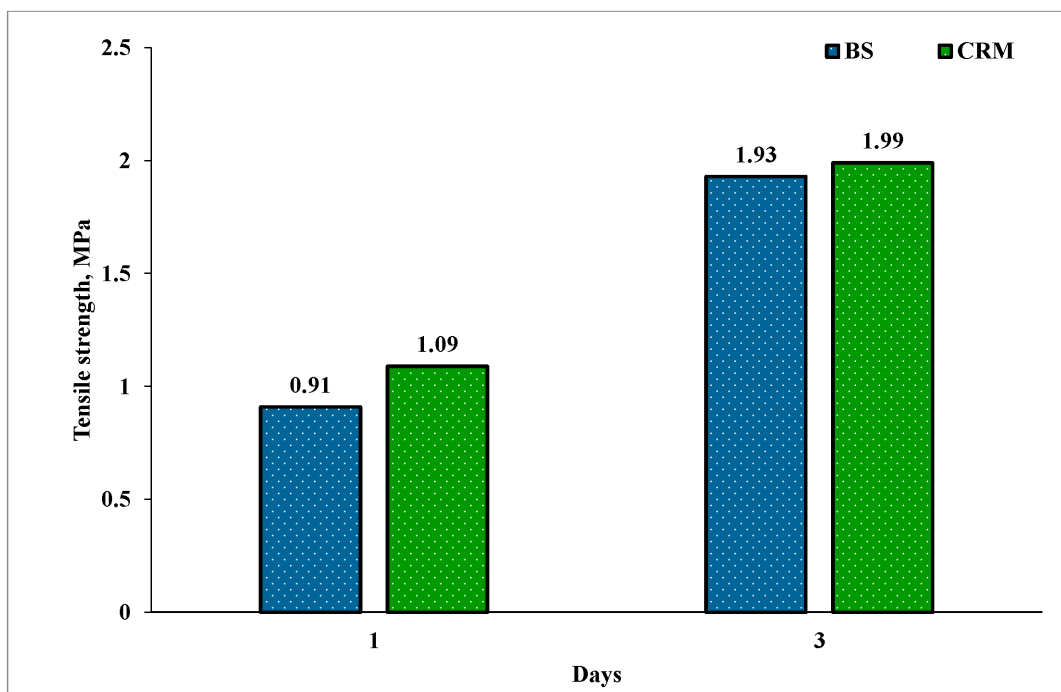


Figure 5—Tensile strength results for the BS and CRM-based slurry.

The consistent performance of the CRM formulation suggests improved bonding capability and crack resistance, even at early curing stages. The narrowing gap in tensile strength between day 1 and day 3 indicates that CRM not only accelerates initial strength gain but also sustains strength development over time. These results support the potential of CRM as a viable partial replacement for cement in LWS applications, contributing to improved tensile behavior and overall mechanical stability.

Porosity

Porosity was assessed through NMR using the transverse relaxation time (T_2) spectrum. The porosity values after 1 and 3 days are illustrated in Fig. 6. Both samples demonstrated a reduction in porosity over the first three days. The neat cement formulation exhibited a decrease in porosity from 59% to 48%, indicating an 18.6% reduction. In contrast, the CRM formulation showed a smaller reduction of 7.8%, decreasing from 64% to 59%. The decline in porosity for both samples underscores the ongoing reactions and the formation of cement hydration products, which gradually fill the pore spaces within the cement sheath over time. This continued hydration process is crucial for improving the mechanical integrity and overall performance of the cement matrix.

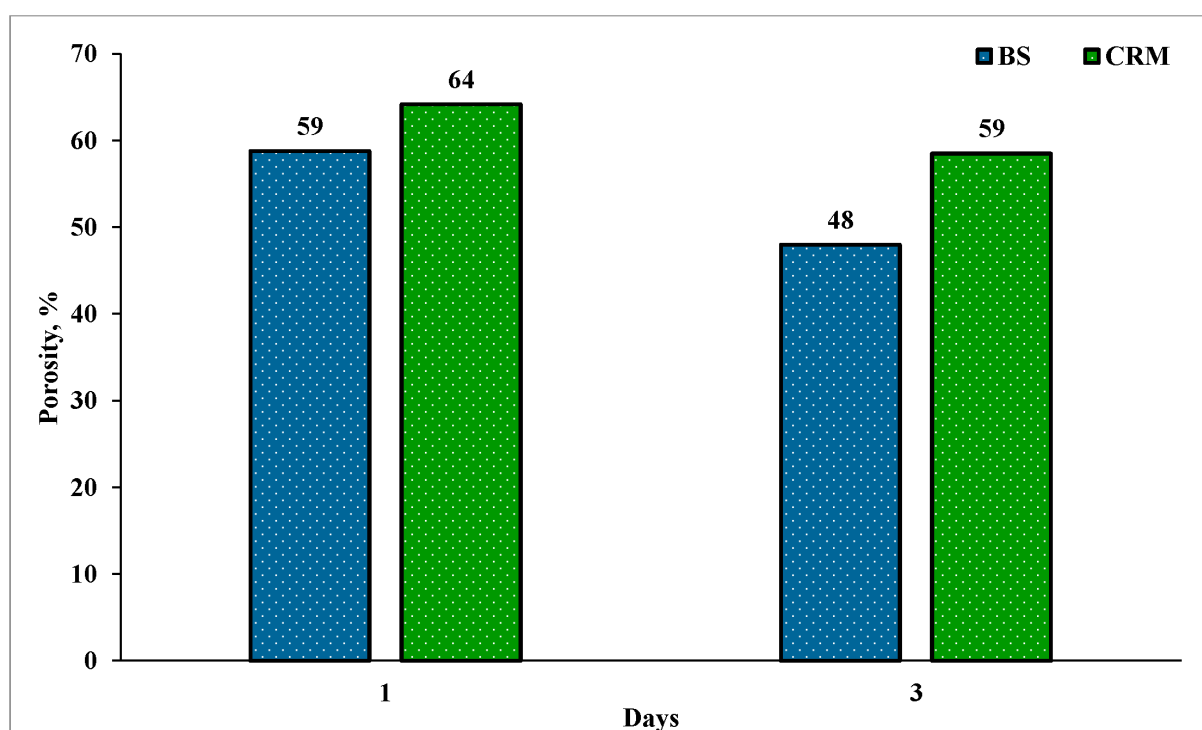


Figure 6—NMR porosity values of the BS and CRM-based samples.

Conclusions and recommendations

This study investigated the potential use of calcinated red mud (CRM) as a partial replacement for cement in lightweight cement slurry (LWS) formulations intended for surface casing applications at 3000 ft depth. Two cement formulations were prepared: a neat cement slurry (base slurry, BS) and a CRM-based slurry. The CRM was obtained by calcining raw red mud at 750 °C for 6 hours. Both the cement and CRM were comprehensively characterized using X-ray diffraction (XRD), X-ray fluorescence (XRF), and scanning electron microscopy (SEM) to assess their mineralogical and morphological properties. Additionally, particle size distribution (PSD) analysis was conducted to evaluate their granulometric characteristics. Mechanical performance and porosity were evaluated at 1 and 3 days of curing.

Key findings are summarized as follows:

- **Compressive Strength:** The BS formulation exhibited higher early compressive strength (29 MPa) compared to CRM (11.4 MPa) on day 1. However, the CRM slurry demonstrated a significant 80% increase in compressive strength by day 3 (20.5 MPa), while BS experienced a 17% decline (24 MPa). This trend suggests that CRM undergoes a slower but sustained hydration process, contributing to improved long-term strength development.
- **Tensile Strength:** CRM-based slurries consistently outperformed BS in tensile strength across both curing intervals. On day 1, CRM exhibited a tensile strength of 1.09 MPa—approximately 20% higher than BS (0.91 MPa). By day 3, CRM reached 1.99 MPa, slightly surpassing BS at 1.93 MPa. The enhanced tensile performance of CRM indicates improved bonding characteristics and crack resistance, even at early stages.
- **Porosity:** Nuclear Magnetic Resonance (NMR) analysis revealed a porosity reduction in both samples from day 1 to day 3. BS porosity decreased from 59% to 48% (an 18.6% reduction), while CRM showed a smaller decline from 64% to 59% (7.8%). These reductions reflect the progressive formation of cement hydration products, gradually filling pore spaces and enhancing the matrix's structural integrity.

The results demonstrate that CRM, when used as a partial cement replacement, exhibits promising mechanical performance, particularly in tensile behavior and long-term strength development. The slower hydration kinetics observed in CRM formulations may contribute to improved durability and stability over extended curing periods. This work represents the first systematic study to evaluate the use of thermally treated (calcinated) red mud in well cementing applications, highlighting its potential as a sustainable and effective material for lightweight cement systems in the oil and gas industry.

Acronyms

RM	Red Mud
CRM	Calcinated Red Mud
BS	Basic Slurry
API	American Petroleum Institute
XRD	X-Ray Diffraction
XRF	X-Ray Fluorescence
SEM	Scanning Electron Microscopy
PSD	Particle Size Distribution
ppg	pound per gallon
pcf	pound per cubic feet
BWOC	By Weight of Cement
BWOB	By Weight of Blend
MPa	Mega Pascal

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